

Sample - > Requirement Analysis Document (RAD)

Project Title: Movie Ticket Booking Database System (Implementation: "CineReserve")

Student Name:

Register Number:

1. Executive Project Summary

CineReserve is an online entertainment platform that allows moviegoers to search for movies, view available showtimes, reserve specific cinema seats, make secure payments, and receive digital tickets. Managing seat reservations in real-time requires a robust transaction network: when a customer books a seat, that specific seat in that specific hall must be locked immediately to prevent duplicate bookings, the payment gateway must be verified, and automated digital barcodes must be sent out.

To manage this safely, CineReserve relies on a Relational Database Management System (RDBMS) like PostgreSQL or Oracle. A relational database is critical because it enforces strict data rules and guarantees

ACID (Atomicity, Consistency, Isolation, Durability) transaction properties. For example:

- **Atomicity** (A): If a user tries to reserve a ticket, the database must *both* confirm the payment and mark those specific seats as Reserved. If the payment fails halfway through, the entire transaction is rolled back so the seat remains available for other moviegoers, and the user is not charged.
- **Consistency** (C): The system prevents invalid states. A ticket price cannot accidentally be saved as negative ($Price \geq 0$), and a booking cannot be completed for a seat that does not physically exist in that cinema hall.

2. Core Business Domains & Features

Based on our analysis of the cinema operations, we have divided the database requirements into four distinct functional modules:

2.1 User & Membership Management

This module handles customer profiles, login credentials, contact info, and loyalty tier rewards.

- **Business Rules:**
 - Every user must provide a unique email address during registration. This email acts as their login credential and cannot be shared.
 - A user is assigned a unique UserID upon registration, alongside their Name, Mobile Number, and a system-generated Registration Date.

- Users can join a "Membership Tier" (such as Standard, Silver, Gold) which tracks their cumulative ticket purchases and offers point multipliers.
- A user can exist in the system without having booked any movie tickets yet.

2.2 Cinema, Movies, & Showtime Management

This module houses the database records for physical cinema theaters, individual screens/halls, movie catalogs, and live showtimes.

- **Business Rules:**

- Movies are organized by "Genres." Every movie must belong to at least one primary Genre (e.g., "Sci-Fi", "Action").
- Physical cinemas are divided into multiple "Halls" (e.g., Screen 1, Screen 2, IMAX Hall). Each Hall contains a specific layout of fixed seats (e.g., Row A, Seat 10).
- A **Showtime** represents a scheduled screening of a specific movie in a specific hall at a specific time (e.g., "Inception on Screen 1 at 7:00 PM").
- The database must enforce the constraint that a single physical Hall cannot host overlapping showtimes ($End_Time_{Showtime1} < Start_Time_{Showtime2}$).

2.3 Booking & Seat Reservation Lifecycle

This module acts as the core transaction engine, capturing which showtimes are chosen, what seats are locked down, and how bookings progress over time.

- **Business Rules:**

- A single booking is identified by a unique BookingID. It is linked to exactly one Customer.
- A booking can contain multiple different seat reservations for the same showtime (e.g., Booking 4 empty seats together). This creates a many-to-many ($M : N$) relationship between Showtime and Seat, which is resolved using a bridging table called Seat Reservations.
- **The Historic Price Rule:** The unit ticket price recorded in Seat Reservations must represent the price *at the exact second of purchase*. If the cinema decides to raise standard weekend ticket prices next week, past purchase receipts must remain unchanged to preserve correct accounting.
- Bookings must support a rigid lifecycle state machine: Selected $\rightarrow$ Pending Payment $\rightarrow$ Confirmed $\rightarrow$ Cancelled $\rightarrow$ Completed.

2.4 Payments & Digital Ticket Operations

This module manages currency validation, digital barcode generation, and scan validations.

- **Business Rules:**

- Each booking has a single Payment profile. A Payment cannot progress without a corresponding valid BookingID ($1 : 1$ relationship).
- A booking status cannot progress to Confirmed until its corresponding Payment status is recorded as Successful.

- Once confirmed, the system must generate a unique digital ticket barcode identifier for each seat booked. The ticket status changes to Completed (or used) only when the scanner at the cinema door validates the barcode.

3. Actor Profiles & System Interactions

To maintain system security, different users must have distinct database access privileges. We define four primary actors:

+-----+	+-----+	+-----+
Actor	----> Read Privileges	----> Write/Edit Rules
+-----+	+-----+	+-----+
Moviegoer	Showtime Listings	Create Bookings
	Personal Tickets	Write Reviews
+-----+	+-----+	+-----+
Cinema Mgr	Box Office Reports	Edit Showtimes
	Seating Layouts	Add Movies
+-----+	+-----+	+-----+
Gate Staff	Ticket Statuses	Scan/Redeem Tix
	Showtime Times	(Cannot see pay)
+-----+	+-----+	+-----+

1. Moviegoer (Customer)

- *What they can Read:* Public movie schedules, showtimes, available seat maps, and their own booking history.
- *What they can Write/Edit:* Their profile info, booking selections, payments, and ratings/reviews on movies they have watched.
- *Safety Boundary:* A Moviegoer can **never** view another user's personal details, credit cards, or ticket history. They cannot change seat pricing or adjust showtime schedules.

2. Cinema Manager

- *What they can Read:* Occupancy statistics, box office revenues, and cinema hall structures.
- *What they can Write/Edit:* Schedule new showtimes, add new movies to the catalog, set seat prices, and declare a screen "Out of Service" for maintenance.
- *Safety Boundary:* Managers cannot view customer credit cards, manually override transactional payment gateways, or delete accounting audit histories.

3. Gate Ticket Scanner (Logistics Staff)

- *What they can Read:* Showtime details, basic movie info, seat assignments, and Ticket Verification Statuses.
- *What they can Write/Edit:* Update digital ticket statuses from Confirmed → Completed upon scan.

- *Safety Boundary*: They cannot view customer financial records, edit seat assignments, cancel showtimes, or alter pricing data.

4. System Admin

- *What they can Read/Write*: Complete *CRUD* (Create, Read, Update, Delete) privileges across all database tables. All Admin transactions are logged to an audit table to prevent internal fraud.

4. Analytical Reporting Requirements

To help CineReserve's management optimize box-office performance, the database must be structurally capable of executing aggregate joins and calculations to produce these three business reports:

- **Report 1: Low-Occupancy Showtime Alerts**

- *Business Value*: Identifies underperforming screenings to help marketing run dynamic discounting (e.g., "cheap Tuesday" promotions).
- *Logical Query Goal*: Find all showtimes occurring in the next 7 days where the percentage of booked seats is less than or equal to a critical threshold ($Occupancy \leq O_t$, where $O_t = 15\%$). The report must list the Movie Title, Showtime Date/Time, Hall Name, total Seat Capacity, and the count of empty seats.

- **Report 2: Cinephile Loyalty Report**

- *Business Value*: Allows the loyalty team to distribute free concessions or premium upgrades to high-spending users.
- *Logical Query Goal*: Retrieve the top 10% of all registered moviegoers ranked by their total successful booking ticket dollar volume over the past 12 months. The report must list the Moviegoer's Name, Email Address, and total amount spent on tickets.

- **Report 3: Genre Box Office Performance Directory**

- *Business Value*: Shows which movie genres generate the most profit, guiding future theater bookings.
- *Logical Query Goal*: For a specified calendar month, calculate the total box-office revenue generated, the total number of tickets sold, and the average movie review rating for each distinct Genre.

5. Non-Functional Requirements & Database Constraints

Our system rules ensure the database is secure, performant, and reliable under real-world conditions:

- **Strict Data Constraints:**

- Rating submissions must be integers between 1 and 5 inclusive ($1 \leq Rating \leq 5$).
- Ticket prices must be strictly positive values ($Price > 0$).
- The booking system must enforce that a seat cannot be double-booked for the exact same showtime.

- **Concurrency Controls:**

- If two users click "Book Seat A10" at the exact same millisecond for the same showtime, the database must use **Row-Level Transaction Locks**. The first user's request will lock the specific row representing that showtime-seat link, process their payment, set the seat status to Reserved, and release the lock. The second user's transaction will then read the updated state (Reserved) and gracefully inform them that the seat is no longer available.

- **Historical Auditing & Safe Deletion:**

- To comply with financial accounting and data privacy regulations, user accounts cannot be permanently hard-deleted. If a user deletes their account, their profile status is updated to Inactive, and personal identifiers (like mobile numbers and billing addresses) are blanked out for privacy compliance. However, their past purchase numbers and tickets must remain in our database to ensure that financial audits and historical sales reports balance perfectly.